ALICE L WALTON SCHOOL OF MEDICINE

Opioid Agonists

Last updated January 29, 2025 · 14 min read · ScholarRx

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Adapted from: Opioids

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Learning Objectives (4) | Versions

After completing this brick, you will be able to:

1. Explain the mechanisms of action of opioid receptor agonists.
2. Classify opioid medications based on their relative potency, efficacy, and binding affinity.
3. Identify the pharmacokinetic properties, routes of administration, and clinical uses of common opioid analgesics.
4. Predict the c[illegible]tients with altered pharmacokinetic parameters due to genetic polymorphism.

CASE CONNECTION

SO is a 52-year-old male patient of Asian descent. He has recently undergone back surgery and is experiencing some chronic severe pain. He was moderately active before the surgery by playing golf four times weekly. He was given an opioid to help manage his pain, but whenever he takes the medication, he feels "too sleepy to do anything but lay on the couch for hours and hours," and the pain without the medication is unbearable.

You are seeing SO in the clinic at a follow-up appointment. He wants to know what you can prescribe to help relieve his pain and help him get back out on the course.

Which medication route will be best for SO? Are there in considerations you might have based on his ethnicity? Consider your answer as you read, and we'll revisit at the end of the brick.

GO TO CONCLUSION ↓

How Do Opioids Work? (LO#1)

Opioid Receptors. Before diving into the mechanism of action and other qualities of opioids, let's take a look a closer look at the receptors.

Opioid receptors are 7-transmembrane G protein-coupled receptors (GPCRs) and they are made up of three primary types, ***Mu* (μ), *kappa* (κ), *and delta* (δ)**. Even though the recep[illegible]ctivated they exert different effects, some [illegible] will be discussed later in the brick.

Opioids are pain-relieving drugs, which are found in peripheral nerves, the brain, the spinal cord, and the gastrointestinal (GI) tract. As opioid receptors found throughout the central and peripheral nervous system are activated, these medications exert their effects. Most of the available opioid medications act primarily on mu-opioid receptors. There are multiple types of receptors serving distinct functions within the central and peripheral nervous systems. The three primary receptors involved in an opioid's analgesic effect are μ, κ, and δ:

- ***Mu-opioid*** receptors in the brain are found in regions that are involved with sensory and affective components of pain, and modulation of many other behaviors (e.g., respiratory drive and reward pathways). See Figure 1 & 2 for specific areas where opioid receptors exert their effects (caudate, putamen, nucleus accumbens, thalamus, hippocampus, amygdala, periaqueductal gray (PAG), medulla and pons, and dorsal horn of the spinal cord).
    - Most of the analgesic properties of opioids are due to activation of the μ opioid receptors. However, when activated, these receptors affect mood and rewarding behavior, as well as altering respiratory, cardiovascular, GI, and neuroendocrine functions. Thus, the activation of the μ receptors can lead to physical dependence, addiction, depression, and euphoria as well as cause respiratory depression and decreased gastric motility. As mu-opioid receptors within the periaqueductal gray area (PAG) are activated, downstream effects provide sensory input to higher centers in the brain and block the sensation of pain.
    - Mu-opioid re[illegible]epression. Significant respiratory depression is not likely to occur at standard analgesic doses of available medications, unless other concomitant factors are present (e.g., asthma, chronic obstructive pulmonary disease (COPD), hypoxia, or hypercapnia). Opioid agonists depress respiratory activity by causing a reduction to intrinsic rhythm generators located in the ventrolateral medulla. This produces a slower rate of breathing and if toxic amounts are ingested the respiratory rate may fall to 3-4 breaths per minute.
    - Opioid agonists reliably decrease gastric motility. Mu-opioid receptors are densely distributed in enteric neurons between the myenteric and submucosal plexuses and on a variety of secretory cells. Commonly, patients treated with opioid agonists develop constipation and changes in bowel function.
- ***Kappa-opioid*** receptors are widely distributed and have prominent roles in a variety of functions including diuresis regulation, food intake, pain perception, and neuroendocrine. Major regions of expression for *kappa-opioid* receptors are also indicated in Figure 1. Additionally, in some instances *kappa-opioid* receptors modulate the effects of *mu-opioid* receptors (caudate, putamen, nucleus accumbens, amygdala, hypothalamus, and pituitary).
    - These receptors can also provide an analgesic effect and modulate the analgesic effect of mu-opioid receptors. Typically, they are not targeted for long-term analgesia because they can produce depression and dysphoria (a sense of unease). When activated, these receptors can cause a prominent diuretic effect as they inhibit the release of oxytocin and antidiuretic hormone (ADH).
- ***Delta-opioid*** receptors are found in fewer areas of the brain, see Figure 1. They are primari[illegible]mood (caudate, putamen, nucleus accumbens, and amygdala).
    - Similar to the other receptors, delta-opioid receptors provide analgesia and can modulate μ receptor activity, they can also produce antidepressant, and anxiolytic effects. The latter effects are related to dopamine modulation.

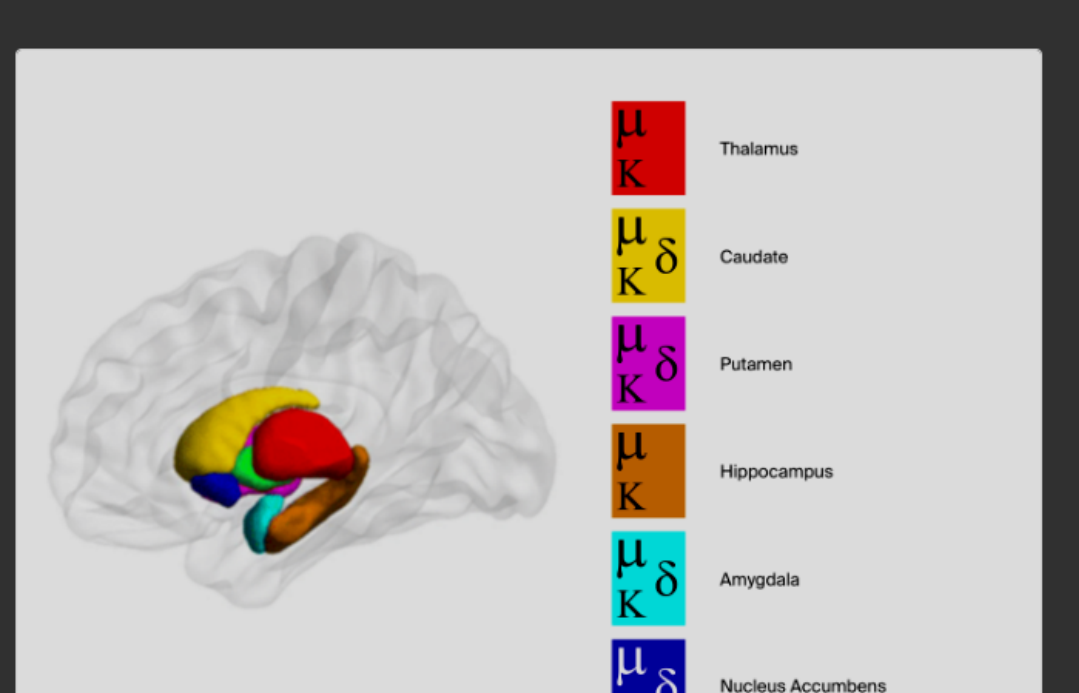

μ κ
Thalamus
μ δ κ
Caudate
μ δ κ
Putamen
μ κ
Hippocampus
μ δ κ
Amygdala
μ δ κ
Nucleus Accumbens
Three primary types of opioid receptors in the CNS: Mu (μ), kappa (κ), and delta (δ)

Figure 1

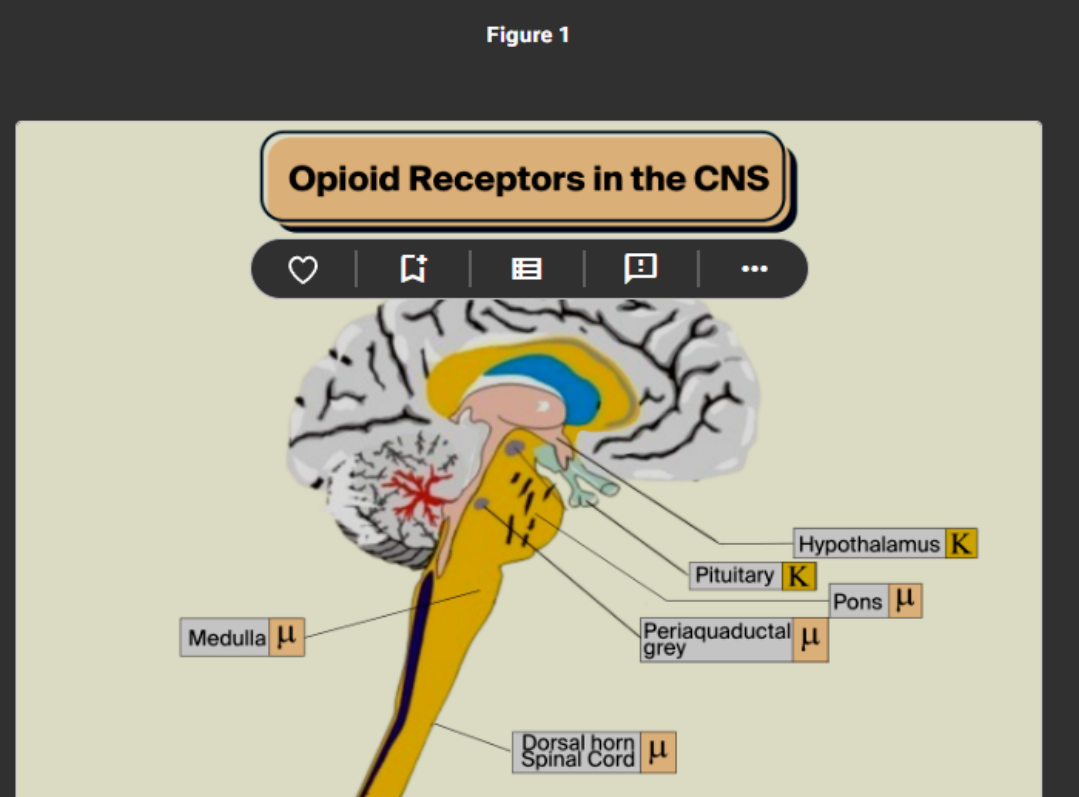


Figure 2

What are the three primary opioid receptors involved in analgesia?

What are full and partial opioid agonists? (LO#2)

Opioids bind to their receptors with different affinities and produce different effects. We can classify opioids depending on whether they act as full agonists, partial agonists, or antagonists. The opioid antagonists will be discussed in another brick.

- **Full agonist**: Opioids that are full agonists completely bind to receptors with high affinity. This leads to wide array of signaling cascades and other physiologic effects. This, in turn, means that opioids that are full agonists have higher efficacy than partial agonists. This is illustrated in Figure 1 below. This category includes most of the familiar opioids, which are listed in Table 1.
- **Partial agonist**: Binds a receptor with high affinity but only induces partial activation of it. At low doses, these drugs act like normal agonists, but at high doses, there is a lesser effect than with full agonists. This has made these drugs useful for analgesia in patients with opioid use disorder, because the patients will get less of a "high" from the drug at high doses. Buprenorphine is the most common partial agonist of the mu-opioid receptor.

A full agonist has 100% efficacy when binding to a receptor, whereas a partial agonist has a lower efficacy (Figure 1).

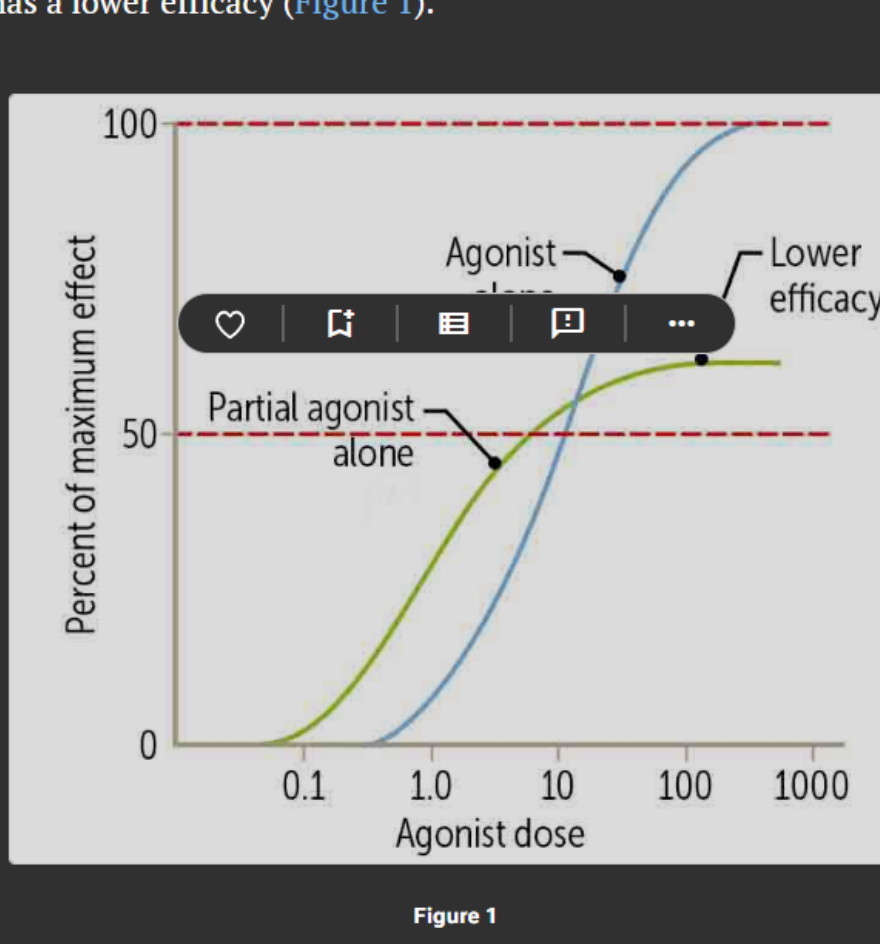


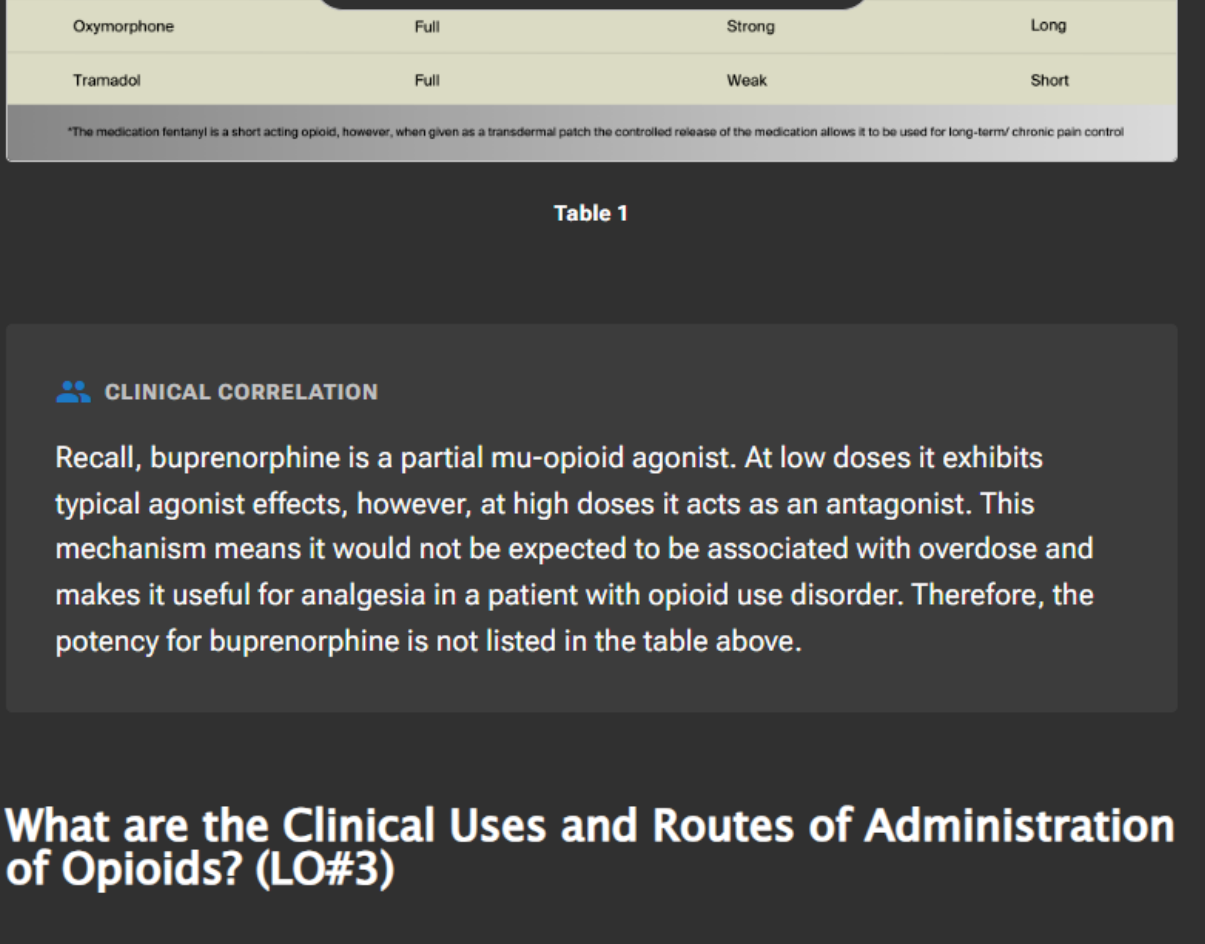
Figure 1

What is the difference between a full and partial opioid agonist?

How do opioids diff[illegible]

Drug potency describes the amount of drug needed for a given effect. With opioid analgesics, we compare different drugs with morphine as a benchmark. For ease, all opioids can be expressed in morphine milligram equivalents (MME), which compares the relative dose of a given opioid to morphine as a standard. Often, for potency of the opioids they are expressed as weak or strong. Refer to Table 1 below for potencies of common opioids.

The duration of action of an opioid agonist refers to how long the receptors remain bound. Or, practically, how long the medication will exert its effects. Different opioids have different durations of action, short or long, and how the drug is given (i.e., dosage form) can play a role in duration as well. Dosage forms of common opioids will be discussed in the next section of this brick. Based on the duration of the drug they can be used for different clinical indications. For instance, short-acting opioids are used for acute pain; long-acting opioids are used for chronic pain management. Table 1 below summarizes the duration of action of common opioid medications.



Opioid	Full or Partial Agonist	Potency	Duration
Buprenorphine	Partial	N/A	Long
Codeine	Full	Weak	Short
Fentanyl	Full	Strong	Short*
Hydromorphone	Full	Strong	Short
Meperidine	Full	Weak	Short
Methadone	Full	Strong	Long
Morphine	Full	Strong	Short
Oxycodone	[illegible]	[illegible]	Short
Oxymorphone	Full	Strong	Long
Tramadol	Full	Weak	Short

*The medication fentanyl is a short acting opioid, however, when given as a transdermal patch the controlled release of the medication allows it to be used for long-term/ chronic pain control

Table 1

CLINICAL CORRELATION

Recall, buprenorphine is a partial mu-opioid agonist. At low doses it exhibits typical agonist effects, however, at high doses it acts as an antagonist. This mechanism means it would not be expected to be associated with overdose and makes it useful for analgesia in a patient with opioid use disorder. Therefore, the potency for buprenorphine is not listed in the table above.

What are the Clinical Uses and Routes of Administration of Opioids? (LO#3)

Opioids are primarily used for **analgesia** in moderate to severe pain. They achieve this because activation of mu-opioid receptors in the PAG produces downstream effects that block the perception of pain. These centrally acting opioids can also produce sedation, an often-unwanted side effect that can be targeted as a clinical use. Codeine is frequently used for its cough suppression. This is achieved as opioid agonists depress the cough reflex within the cough center of the medulla. In patients with opioid use disorder buprenorphine and methadone can be used. Buprenorphine, as outlined above in this brick, is a partial agonist, at low[illegible]nd high doses acting as an antagonist to re[illegible]e is a full agonist with an ultra-long duration of action, making it useful for a sustained effect in the patient with opioid use disorder.

Opioids are administered through a variety of routes, and based on the route through which they are given it can alter the pharmacokinetic properties.

- **Oral (PO)**: Many of the common opioids are available in this route, and this route is best for mild, moderate, and chronic pain. It is convenient and non-invasive for patients. However, from a pharmacokinetic standpoint, it subjects the drug to first pass metabolism (FPM) by the liver, possible stability issues in the GI tract, and potentially lower bioavailability.
- **Intravenous (IV)**: This is the preferred route for acute severe pain. It provides a quicker onset of action, avoids FPM, and most often provides a higher bioavailability than the PO route.
- **Intramuscular (IM)**: This is the least common route for the opioids, but it can be helpful in acute situations where IV access is not yet obtained/ possible. FPM is avoided with this route, but it has a slower onset than IV. Bioavailability of the drug is generally good in this route, though generally not as high as the IV route.
- **Transdermal**: Fentanyl transdermal patches are the most common opioid utilized in this route, and they are approved for use in chronic/sustained pain. The slow rate of absorption through the skin provides a pharmacokinetic advantage by increasing the duration of action of the medication. This route provides ease of use and stable blood levels. It is especially useful for cancer-related pain treatment.

CLINICAL CORREL[illegible]

Very potent opioid medications, like Fentanyl and hydromorphone, can produce a strong sedation effect through their actions on the CNS. While in the outpatient setting this side effect can be concerning, this 'side-effect' is often targeted for inpatients. For example, continuous infusions (i.e., drips) of fentanyl are frequently utilized immediately following endotracheal intubation to target analgesia and sedation, or analgosedation. This medication use strategy can maintain patient comfort and compliance with mechanical ventilation.

Which opioid administration route is best for mild/ moderate pain?

What about patients with altered pharmacokinetic parameters due to genetic polymorphism? (LO#4)

It is important to keep in mind that some opioids are metabolized in the liver by the cytochrome P450 (CYP450) enzyme system. Specifically, many opioids are metabolized by C[illegible] hundreds of drug-drug interactions that may occur. It is important to be aware of medications that inhibit or induce the CYP450 system, because it has the potential to increase or decrease opioid levels. These altered levels have the potential to enhance the side effect profile (e.g., respiratory depression, sedation, constipation) of these drugs.

In children less than 12, the opioid codeine has an FDA black box warning due the risk of respiratory depression and death. Once ingested, codeine is metabolized by CYP 2D6 into morphine before undergoing further metabolism. Many children can have a genetic polymorphism in CYP 2D6 that predisposes them to ultra rapid metabolism with this enzyme. This can cause dangerously high levels that cause severe respiratory depression and death.

CLINICAL CORRELATION

Recent evidence shows that opioid receptor function has increasing complexity with large genetic variations. This can manifest as highly varied responses to the same opioid medications among different ethnic groups. This is known as altered pharmacogenomics. For example, a single nucleotide polymorphism for the 2D6 enzyme can make an individual an ultrapid (UR), normal, intermediate, or poor metabolizer. A UR metabolizer clears medication from the body very quickly and will experience significantly shorter durations of action with opioids. This UR polymorphism can be common (up to 30%) in individuals of Middle Eastern and North African descent. On the other hand, poor metabolizers will have altered pharmacokinetics that will prolong the duration of action of opioids. This means a single dose of a medication could last two to three times as long as expected and could lead to an unintended overdose. Poor metabolizers are most common among individuals of [illegible]

Ultrarapid metabolizers are most common in which ethnic group(s)?

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to SO, which medication route will be best for him, and what considerations might you have based on ethnicity?

Given our brief clinical vignette of SO, we know that we need to help manage his chronic severe pain following back surgery. SO was prescribed an opioid following the surgery, so checking into what he got will be important. A PO medication will likely be sufficient to control his pain, but the overly sedative effects are concerning for SO. Given that he is of Asian descent, there is a strong possibility that he could be a poor metabolizer and experiencing enhanced side effects for that reason. You could consider extending the time between doses or lowering the dose to decrease the [illegible]f.

Summary

- Opioid receptors are GPCRs made up of three primary types: Mu (μ), kappa (κ), and delta (δ).
- These receptors are distributed throughout the body in the peripheral and central nervous system, exerting different effects when activated.
- Opioids are analgesic (pain-relieving) drugs that act on opioid receptors, with most of the available opioid medications acting primarily on mu-opioid receptors.
- Opioids can be classified as full agonists, partial agonists, or antagonists, depending on their binding affinity.
- Opioids are primarily used for analgesia in moderate to severe pain, but they can also produce sedation, and/ or suppress cough.
- The pharmacokinetic properties of opioids can be altered by the route of administration, drug-drug interactions, and genetic variations in opioid receptor function.
- Opioids metabolized by the CYP450 enzyme system, particularly CYP 3A4 and 2D6, can have significant drug-drug interactions.
- In children under 12, codeine poses a high risk of respiratory depression and death due to genetic polymorphisms in CYP 2D6 that lead to ultra-rapid metabolism.

Review Questions

1. Which opioid rece[illegible]e analgesic effects?

A. ○ Delta
B. ○ Kappa
C. ● Mu
D. ○ Omega

Hide Explanation

The correct answer is Mu-opioid receptors (C). While delta-opioid receptors (A) and kappa-opioid receptors (B) may have a role in analgesia, the primary analgesic effect is produced as the mu-opioid receptor agonists produce the pain block sensations. Omega (D) is not a an opioid receptor type and makes it an incorrect choice.

2. Comparing full and partial opioid agonists, they both bind receptors with high affinity, but the full agonist will have higher _______ than partial agonists?

A. ○ Absorption
B. ○ Duration
C. ● Efficacy
D. ○ Excretion

Hide Explanation

The correct answer is efficacy (C). Full agonists have high affinity for the opioid receptors and produce a large variety of effects. This means they have higher efficacy compared to partial agonists. The absorption (A), duration (B), and excretion (D) is not affected based on whether the opioid is a full or partial opioid agonist making these options incorrect.

3. Which route of opioid administration avoids first pass metabolism and provides the quickest onset?

A. ○ Oral
B. ● IV
C. ○ IM
D. ○ Transdermal

Hide Explanation

The correct answer is Intravenous (B). The IV route avoids first pass metabolism (FPM) and provides a quicker onset of action. Oral (A) is subject to FPM and has a slower onset. Intram[illegible]set than IV. Transdermal (D) has a slow rate of absorption and is used for sustained pain management.

4. A patient of _______ ethnic descent would you be most likely to be a CYP 2D6 poor metabolizer.

A. ● Asian
B. ○ Caucasian
C. ○ Middle Eastern
D. ○ North African

Hide Explanation

The correct answer is Asian (A). Individuals of Asian descent are most likely (up to 35%) to have a genetic polymorphism predisposing them to be poor opioid metabolizers. This can alter the pharmacokinetics of the drug and prolong the duration, presenting as enhanced side effect profiles of opioids. Individuals of Middle Eastern (C) and North African (D) descent are most likely to have altered pharmacokinetic profiles that lead to enhanced/ rapid clearance of opioids, which decreases the duration of action. Individuals of Caucasian (B) descent can experience poor or rapid metabolism of opioids; however, it is not nearly as common as the other ethnic groups.

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How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Explain the mechanisms of action of opioid receptor agonists.~~
02 ☑ ~~Classify opioid medications based on their relative potency, efficacy, and binding affinity.~~
03 ☑ ~~Identify the pharmacokinetic properties, routes of administration, and clinical uses of common opioid analgesics.~~
04 ☑ ~~Predict the clinical implications of opioids in the patients with altered pharmacokinetic parameters due to genetic polymorphism.~~

Objective Confidence: 4/4